

ULTRASONIC DISTANCE METER Using Raspberry Pi 2

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Ultrasonic ranging module HC-SR04 provides 2cm to 400cm non-contact distance measurement range. Its accuracy can reach $\pm 3\text{mm}$ with a cycle period of 50ms, which is the minimum time delay recommended before measurements are taken. The module includes ultrasonic transmitter, receiver and control circuit.

Raspberry Pi 2 is a second-generation Raspberry Pi board, which was released in February 2015. It is a Linux based computer with a powerful Broadcom BCM2836 ARMv7 900MHz quad-core processor and 1GB SDRAM. It runs on 5V DC at 200mA and has 40 GPIO pins, along with HDMI, micro-SD card slot, Ethernet, USB 2.0 and 3.5 audio (out) ports.



Fig. 1: HC-SR04 ultrasonic sensor module

Circuit and Working

Components required for this project are Raspberry Pi (or Raspi) 2, HC-SR04 module, 330-ohm and 470-ohm resistors, and some jumper wires.

Working of HC-SR04 ultrasonic sensor is fairly simple. Circuit diagram of Raspi with HC-SR04 is shown in Fig. 3. The HC-SR04 module and its working are shown

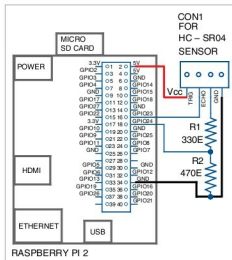


Fig. 3: Connection of HC-SR04 module with Raspi

in Figs 1 and 2, respectively.

When the processor gives a high signal to trig pin of the sensor, the sensor emits out eight-cycle sonic bursts at 40kHz. After sending the sonic burst, the sensor sets echo to high till the sonic burst returns to the sensor—after reflection from the object. Hence, length of this pulse is proportional to how far the object is.

Now, distance covered by the pulse is twice the distance that has to be measured, and speed of sound is roughly 340 metres per second. Calculations can be carried out using the following relationship:

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}, \text{ or}$$

$$340 = \frac{2 \times \text{Distance}}{\text{Time}}, \text{ or}$$

$$\text{Distance} = 170 \times \text{Time}$$

Hence, distance between the object and the sensor in metres can be determined. To calculate in centimetres, multiply it by 100.

Connect HC-SR04 to Raspi as indicated below (also refer Fig. 3):

HC-SR04	Raspi pins
V _{CC} (+5V)	Pin 2 (V _{CC})
TRIG	Pin 16 (GPIO 23)
ECHO	Pin 18 (GPIO 24)
GND	Pin 34 (GND)

Resistors R1 and R2 are used to balance out the output as the sensor's high signal is at +5V, whereas Raspi works on +3.3V level. Hence, the voltage divider helps in reducing the output voltage level.

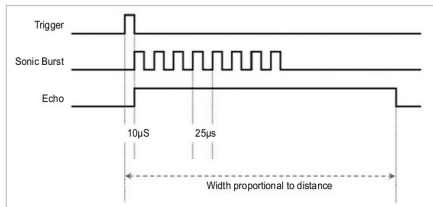


Fig. 2: Working of HC-SR04